



AI Fairness CIO

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FDK™ Toolkit, Governance Domain Fairness Audit

External Validation Report: Folktables ACSPublicCoverage Dataset

Seventh and Final Domain Real-World Validation, Completing All FDK Toolkit Domains

Dataset: Folktables ACSPublicCoverage (Ding, Hardt, Miller & Schmidt, NeurIPS 2021), California 2018 ACS PUMS extract, 138,554 individuals

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Abstract

This report documents the validation of the FDK™ Toolkit's Governance fairness-audit pipeline against a real, publicly available benchmark dataset, Folktables' ACSPublicCoverage task, completing real-dataset validation across all seven FDK Toolkit domains (following Justice, Education, Finance, Health, Hiring, and Business). As with Health and Hiring, Governance's no-code dropdown UX did not exist before this validation cycle and was built from scratch. This validation also independently reproduced, on real data, two defect patterns previously found and fixed in other domains: a statistical-fallback binary-detection gap (found in every domain so far) and, notably, the same all-or-nothing subgroup-rejection defect first identified and fixed in the Justice domain's COMPAS validation, here triggered by a genuinely real 17-person Alaska Native subgroup among 138,554 records. Two further defects specific to this domain's own composite scoring logic were found and fixed: the headline fairness assessment was driven by a composite index diluted with non-fairness components (model accuracy and a transparency measure) rather than the domain's own bias-specific score, and that bias score itself conflated absolute model performance with actual cross-group disparity. With all defects resolved, the Governance pipeline produced complete, internally consistent 27-metric audits across two protected-attribute cuts, race and sex, on the same underlying population, again reproducing the pattern found in every other domain validated in this project: one attribute clean, the other showing substantial disparity.

1. Background and Purpose

Following Business's validation as the sixth domain, Governance was selected as the seventh and final domain, continuing the project's established file-by-file review methodology. Governance's four files (`fdk_governance.py`, `fdk_governance_pipeline.py`, `auto_confirm_governance.html`, `upload_governance.html`) were found to closely mirror Business's own starting structure, including the same class of statistical-fallback defect, though `upload_governance.html` notably did not repeat Business's hardcoded-URL or `display:none`-ancestor defects, having already used a relative path and a clean field structure from the outset.

The Folktables ACSPublicCoverage dataset was selected for this validation: a real dataset derived directly from US Census Bureau American Community Survey (ACS) microdata, purpose-built for fairness research as a documented successor to the UCI Adult dataset, and predicting whether a low-income individual is covered by public health insurance, a genuine government-benefit outcome, directly matching this domain's own keyword conventions (`benefit_approval`, `program_eligibility`, `resource_access`). Its use here serves the same purpose as the Justice domain's COMPAS validation and every subsequent domain's real-dataset work: confirming the FDK™ Toolkit's own correctness against real, messy data.

2. Dataset Description

Property	Value
Total records	138,554 individuals (California, 2018 ACS PUMS 1-year extract)
Extraction method	Folktables Python package (Ding et al., NeurIPS 2021), one-time extraction to a static CSV, the ongoing validation workflow uses this CSV like every other dataset in this project
Group columns used	race (RAC1P, 9 categories) and sex (2 categories)
Ground-truth column	PUBCOV (covered by public health insurance, binary)
Prediction column	predicted_coverage (toy LogisticRegression classifier, class_weight='balanced')
Probability column	predicted_probability (toy classifier output)
Race group sizes	White: 78,600 · Some Other Race: 21,932 · Asian: 20,835 · Black: 8,000 · Two or More Races: 6,955 · American Indian: 1,245 · Native Hawaiian/Pacific Islander: 526 · American Indian/Alaska Native (other): 444 · Alaska Native: 17
Sex group sizes	Female: 77,446 · Male: 61,108

One property of this real dataset is directly relevant to Finding 4 below: the Alaska Native group (n=17) falls below the toolkit's minimum-subgroup-size threshold of 20 samples used elsewhere in this project ;the same situation, with a near-identical group size, as the Native American group (n=18) that originally motivated the Justice domain's subgroup soft-exclusion fix.

3. Findings From Real-Dataset Validation

Finding 1, Statistical fallback accepted any two-valued column as binary

detect_governance_column_mappings already contained a correct helper, `_is_binary_column`, verifying that a candidate column's values are a genuine subset of $\{0,1\}$, and this helper was already used correctly throughout the function's keyword-matched detection layer. The function's statistical fallback, however, checked only whether a column had exactly two unique values, with no verification of what those values actually were, the same defect pattern found and fixed in Finance, Health, Hiring, and Business.

Remediation: both fallback branches now call the existing `_is_binary_column` helper directly.

Finding 2, Missing independent binary-validation layer in the metrics pipeline

`fdk_governance_pipeline.py`'s `validate_dataframe` function coerced non-numeric `y_true/y_pred` columns to numeric where possible, but never verified the result was genuinely binary. A column with three or more distinct values, or a $\{1,2\}$ -coded column, would have passed undetected.

Remediation: an independent binary-validation check was added, verifying strict $\{0,1\}$ membership, consistent with the three-layer defense pattern (detection-time check, confirmation-page sanitization, pipeline-level independent validation) used across every FDK domain in this project.

Finding 3, No manual-override UX existed anywhere in the domain

As with Health and Hiring, no dropdown-based override UX existed for Governance prior to this work: the confirmation page was static and read-only, and its hidden confirmation-page form inputs were submitted but never read by the backend route.

Remediation: the same no-code dropdown-support functions built for the four prior domains (`_describe_column`, `_candidate_columns`, `build_column_options`) were constructed for Governance and wired into both the confirmation page (four real, filtered, pre-selected dropdowns) and the backend route (override-reading, re-validated against each role's candidate list).

Finding 4, All-or-nothing subgroup rejection (identical defect to Justice's Finding 1)

The initial validation run against the real race-cut data rejected the dataset in its entirety: `metrics_calculated`: 0, with the error citing the Alaska Native subgroup's 17 samples as below the 20-sample threshold. This is the exact same defect independently identified and fixed in the Justice domain's COMPAS validation, a single small-but-real demographic group (here, 0.01% of 138,554 records) was sufficient to block analysis of the other 99.99% entirely, including groups with very strong statistical power (White, $n=78,600$).

Remediation: the identical soft-exclusion policy verified in Justice was applied to Governance: groups below the 20-sample threshold are excluded from the analysis, with the exclusion and reason recorded transparently in the output (`excluded_groups`), and the audit proceeds on the remaining adequately-sized groups. A hard rejection is retained only for the genuinely degenerate case of fewer than two groups remaining after exclusion.

Verification: confirmed directly against the real dataset, `metrics_calculated`: 0 before the fix, `metrics_calculated`: 27 (with `excluded_groups`: {"Alaska Native": 17} and `sample_size`: 138,537) after.

Finding 5, Headline assessment driven by a diluted composite index rather than the domain's own bias score

`assess_governance_fairness` generated the headline `overall_assessment` text using only `composite_governance_fairness_index` (CGFI), a metric computed as 40% actual fairness metrics, 30% raw model accuracy, 20% a transparency measure, and 10% data-integrity, none of which beyond the first component measures cross-group disparity. `composite_bias_score` (built purely from genuine disparity metrics, consistent with every other FDK domain's convention) and `critical_issues` (a list of specific fairness concerns) were both computed correctly but never factored into the headline text. On the first real race-cut run, this produced a report stating "GOOD - Generally fair with minor improvements needed" directly above a `critical_issues` list in which all four possible warnings had fired, including "Unacceptable performance gaps for disadvantaged constituent groups", a direct self-contradiction within the same report.

Remediation: `assess_governance_fairness` was changed to use `composite_bias_score` directly, with the same LOW_BIAS/MEDIUM_BIAS/HIGH_BIAS three-tier convention and thresholds already used consistently across Finance, Health, Hiring, and Business.

Verification: confirmed against the real race-cut data, the corrected run reads "HIGH_BIAS - Significant policy equity concerns requiring urgent review", now consistent with both `composite_bias_score` (0.79 at the time) and all four `critical_issues`.

Finding 6, composite_bias_score conflated absolute model performance with cross-group disparity

Even after Finding 5's fix, the sex-cut audit's `composite_bias_score` (0.127) crossed into HIGH_BIAS territory despite `statistical_parity_difference` (0.020) and `disparate_impact_ratio` (0.947) both indicating negligible disparity between Male and Female. Root-caused to the score's `worst_case_subgroup_performance` component: this measures the absolute performance (mean of accuracy, precision, recall, and F1) of whichever single group scores worst, not a gap between groups. A model that performs identically, if mediocrately, across every group, with zero actual disparity, would still be scored as biased by this term, since it cannot distinguish "treated worse than others" from "equally weak for everyone."

Remediation: `worst_case_subgroup_performance` was removed from the `composite_bias_score` formula, with its former weight redistributed proportionally across the four remaining, genuinely disparity-based components, so the score reflects cross-group disparity only.

Verification: confirmed against both real cuts, the sex cut's score dropped from 0.127 (HIGH_BIAS) to 0.071 (MEDIUM_BIAS), correctly reflecting its genuinely small disparity; the race cut's score rose from 0.787 to 0.838, remaining correctly classified as HIGH_BIAS, since its actual disparity metrics were severe independent of this component.

Verification of fixes

Check	Before fix	After fix
Statistical fallback accepts non-binary two-valued columns	Yes (undetected)	No , correctly rejected
Pipeline independently validates y_true/y_pred are binary	No such check existed	Present, flags non-binary columns
Confirmation-page column overrides	Static display, no override capability	Four real, filtered dropdowns; overrides read and re-validated server-side
Race-cut audit (n=17 Alaska Native subgroup present)	metrics_calculated: 0 (entire dataset rejected)	metrics_calculated: 27; excluded_groups: {"Alaska Native": 17}; sample_size: 138,537
Headline assessment vs. critical_issues consistency	"GOOD" headline despite 4/4 critical_issues firing	"HIGH_BIAS" headline, consistent with critical_issues and composite_bias_score
composite_bias_score, sex cut (negligible actual disparity)	0.127 (HIGH_BIAS , driven by absolute performance, not disparity)	0.071 (MEDIUM_BIAS , reflects genuine small disparity only)

4. Fairness Findings on the Folktables ACSPublicCoverage Dataset

With all six defects resolved, the Governance pipeline produced two complete, independently verified 27-metric audits: one per protected-attribute cut, on the same 138,554-individual population and the same toy prediction model.

Cut 1: race

Metric	Value	Interpretation
disparate_impact_ratio	0.609	Below the conventional 0.80 four-fifths threshold
statistical_parity_difference	0.198	A 19.8 percentage-point gap between the highest and lowest group
selection_rates_by_group (Black)	0.505	Highest predicted-coverage rate of the eight analysed groups (Alaska Native excluded, n=17)
selection_rates_by_group (Asian)	0.307	Lowest predicted-coverage rate
composite_bias_score	0.838	Toolkit's own classification: HIGH_BIAS, the most severe reading found in any domain validated in this project

Cut 2: sex

Metric	Value	Interpretation
disparate_impact_ratio	0.947	Comfortably clear of the 0.80 threshold
statistical_parity_difference	0.020	A 2.0 percentage-point gap between Male and Female
selection_rates_by_group (Female)	0.365	
selection_rates_by_group (Male)	0.385	
composite_bias_score	0.071	Toolkit's own classification: MEDIUM_BIAS

These findings replicate, for the seventh consecutive time across every domain validated in this project, the same core pattern: auditing only one protected attribute on a given dataset does not license a general conclusion about that dataset or model's fairness. Race shows the single most severe disparity found anywhere in this project's validation work (composite_bias_score 0.838); sex, on the identical population and model, shows only moderate disparity (0.071). This particular contrast is also directionally consistent with this dataset's treatment in the academic fairness literature, where race, sex, and age are all established as meaningful protected attributes for this exact task.

5. Scope Limitations of This Validation Run

- The dataset contains no native model-decision column; a toy LogisticRegression classifier was trained to derive predicted outcomes. These fairness findings describe the toy model's behaviour, not any real government eligibility-determination system, and do not represent a claim about current US public health insurance administration.
- This validation uses one state's (California) 2018 ACS data; Folktables itself documents substantial state-to-state and year-to-year variation in fairness properties for this exact task, so these findings should not be generalised to other states or years without further testing.
- The Alaska Native group (n=17) was excluded from the race-cut analysis per the soft-exclusion policy; this is a real, small population in California's ACS sample, not a data quality artifact, and its exclusion means the race-cut findings do not speak to outcomes for that specific group.
- The Governance no-code UX (dropdowns and override handling) is new construction verified against a single real dataset so far, not the breadth of edge cases the longer-established Finance and Health domains have now seen.

- `composite_bias_score`'s reformulation (Finding 6) changes its numeric scale from the version used in this project's very first runs; historical Governance reports generated before this fix are not directly comparable to reports generated after it.

6. Conclusion

The Governance domain fairness-audit pipeline and its no-code user experience, the latter built from scratch during this validation cycle, have been successfully validated against a real, independently collected, extensively cited benchmark dataset. Six genuine defects were identified, root-caused, fixed, and verified, including one, the all-or-nothing subgroup rejection, that is the exact same defect independently found and fixed during the Justice domain's own COMPAS validation, now confirmed as a recurring, cross-domain pattern rather than a one-off issue, and two others (the diluted headline assessment and the performance/disparity conflation in `composite_bias_score`) unique to this domain's own scoring logic. The pipeline's resulting fairness findings reproduce, for the seventh time in this project, the core pattern that a single dataset and model can be simultaneously fair on one protected attribute and severely unfair on another. With Governance now validated, all seven FDK™ Toolkit domains (Justice, Education, Finance, Health, Hiring, Business, and Governance) have completed real-dataset validation under this same methodology.

Appendix, Reference Materials

Dataset: `governance_acspubliccoverage_raw.csv` (Folktables ACSPublicCoverage, Ding, Hardt, Miller & Schmidt, NeurIPS 2021; California, 2018 ACS PUMS 1-year extract)

Modified source files: `fdk_governance.py` (statistical-fallback fix; `_describe_column`, `_candidate_columns`, `build_column_options`; sanitization and override handling); `fdk_governance_pipeline.py` (independent binary-validation check; soft-exclusion policy for small subgroups; `assess_governance_fairness` switched to `composite_bias_score`; `composite_bias_score` reformulated to remove `worst_case_subgroup_performance`); `auto_confirm_governance.html` (static display replaced with filtered dropdowns); `upload_governance.html` (target-column dropdown added)

Validation reports referenced: `governance_audit_report_20260723_134455.json` (race cut, pre-Finding-5/6 fixes); `governance_audit_report_20260723_140100.json` (race cut, post-Finding-5 fix); `governance_audit_report_20260723_140212.json` (sex cut, post-Finding-5 fix, pre-Finding-6 fix); `governance_audit_report_20260723_141046.json` (sex cut, fully resolved); `governance_audit_report_20260723_141118.json` (race cut, fully resolved)

Related reports: `Justice_COMPAS_External_Validation_Report.docx` (source of the Finding 4 precedent and this report's format); `Finance_API_External_Validation_Report.docx`; `Health_API_External_Validation_Report.docx`; `Hiring_API_External_Validation_Report.docx`; `Business_API_External_Validation_Report.docx`